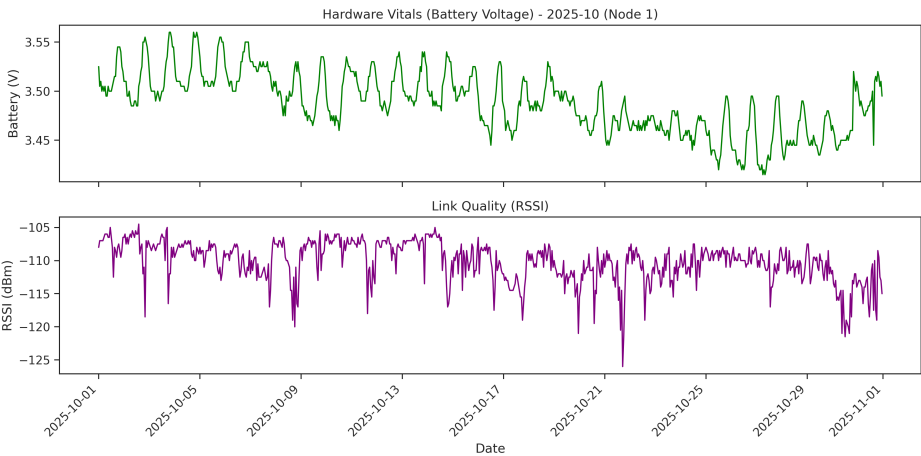


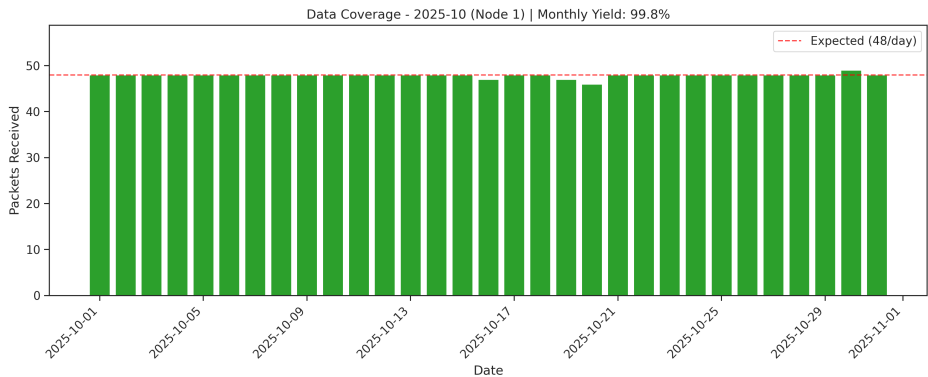
Executive Summary - 2025-10 (Node 1)

| Metric                   | Value                    |
|--------------------------|--------------------------|
| Total Packets            | 1485                     |
| Uptime Yield             | 99.8%                    |
| Temp (Min / Mean / Max)  | 5.6 / 12.4 / 20.8 °C     |
| VWC (Min / Mean / Max)   | 2102.9 / 2270.5 / 2732.5 |
| Battery Drop (Max - Min) | 0.180 V                  |
| Battery Absolute Min     | 3.380 V                  |

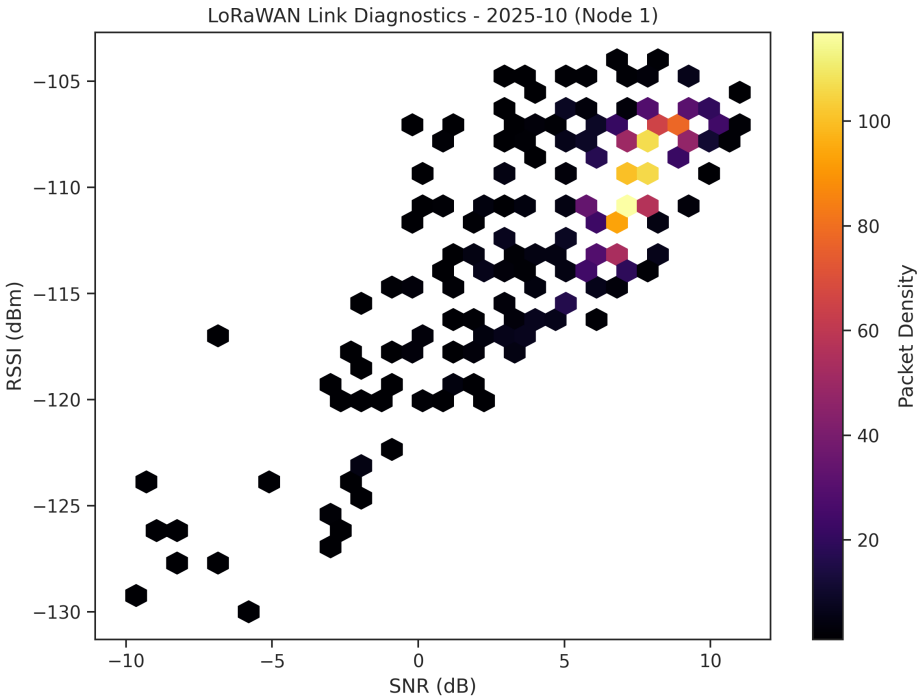
Overview of monthly data yield, temperature, raw moisture boundaries, and battery voltage range.



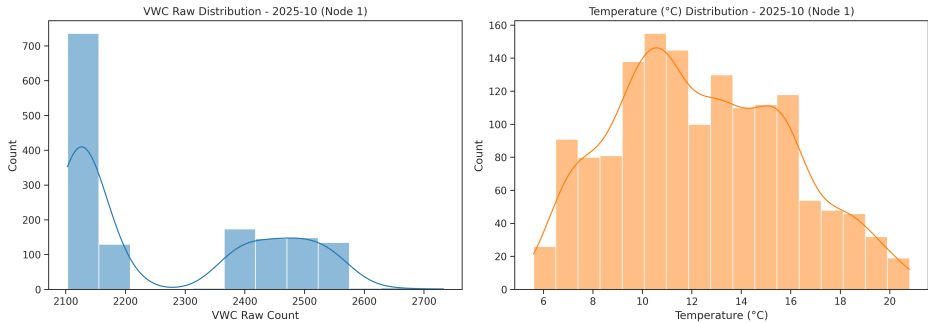
Monitors battery voltage depletion alongside wireless Link Quality (RSSI) to verify hardware stability.



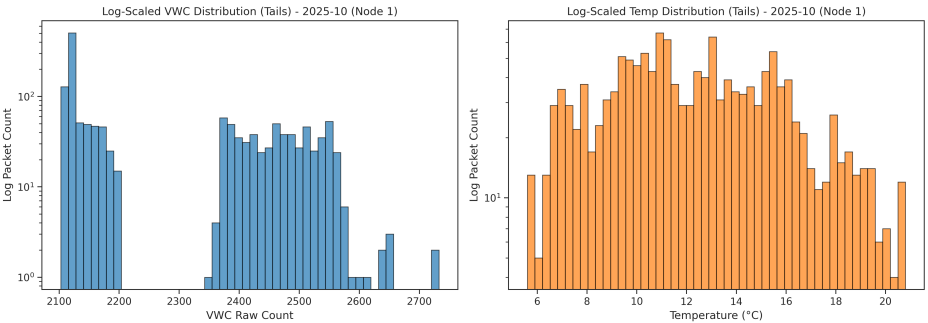
Daily packet reception count. The red line represents the target expected baseline of 48 packets/day (one every 30 mins).



Signal-to-Noise Ratio (SNR) vs. Received Signal Strength (RSSI) density heatmap to assess connection quality and canopy wetness attenuation.

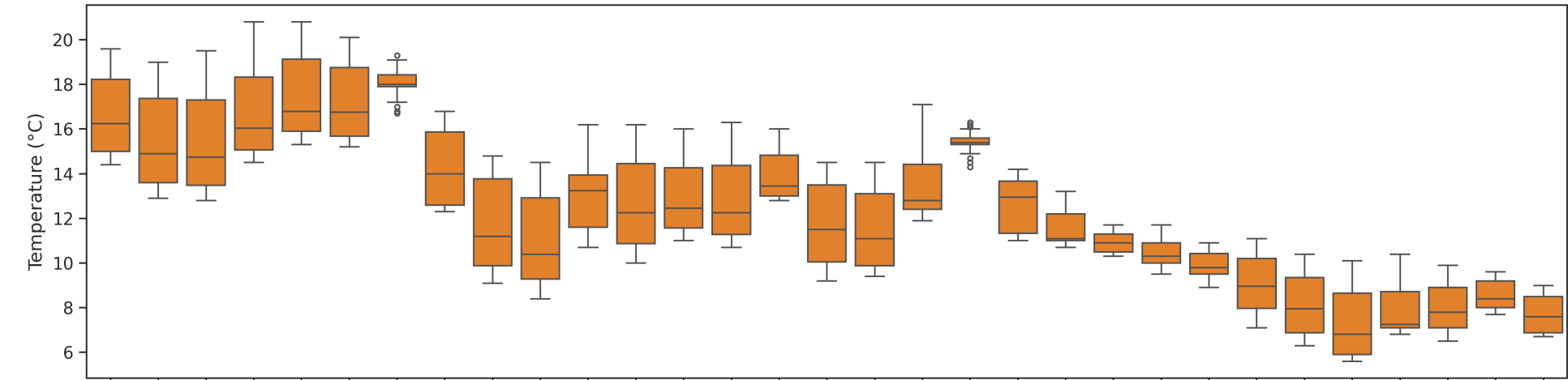


Standard probability distribution of raw VWC counts and temperature readings showing main operating modes.

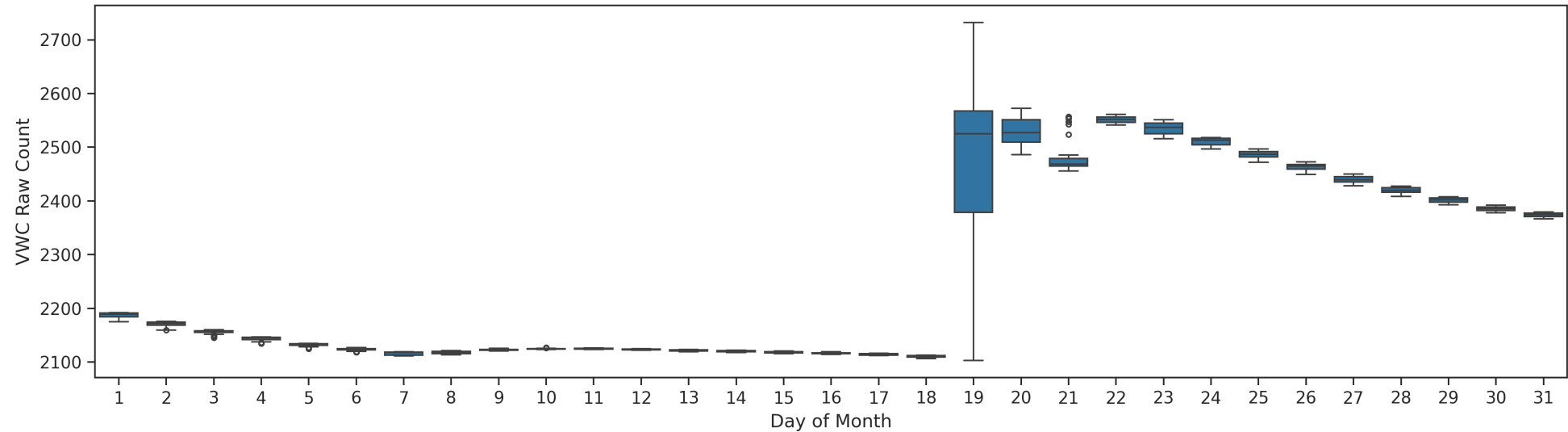


Log-scaled histograms designed to isolate rare anomalies, electrical spikes, and extreme weather events in the distribution tails.

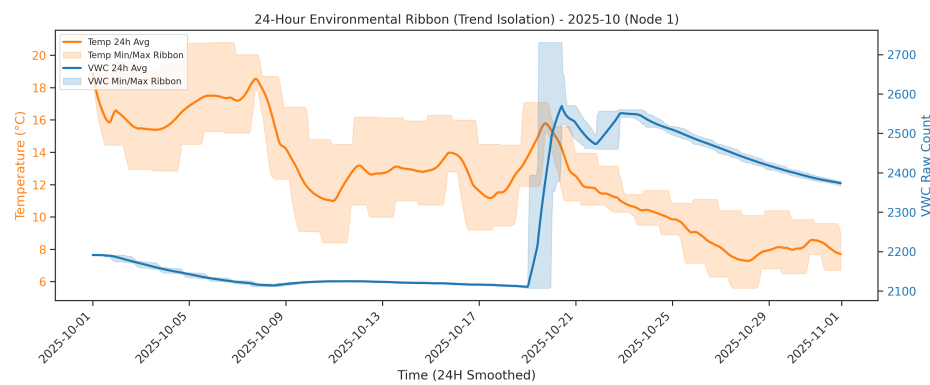
Daily Temperature Distributions - 2025-10 (Node 1)



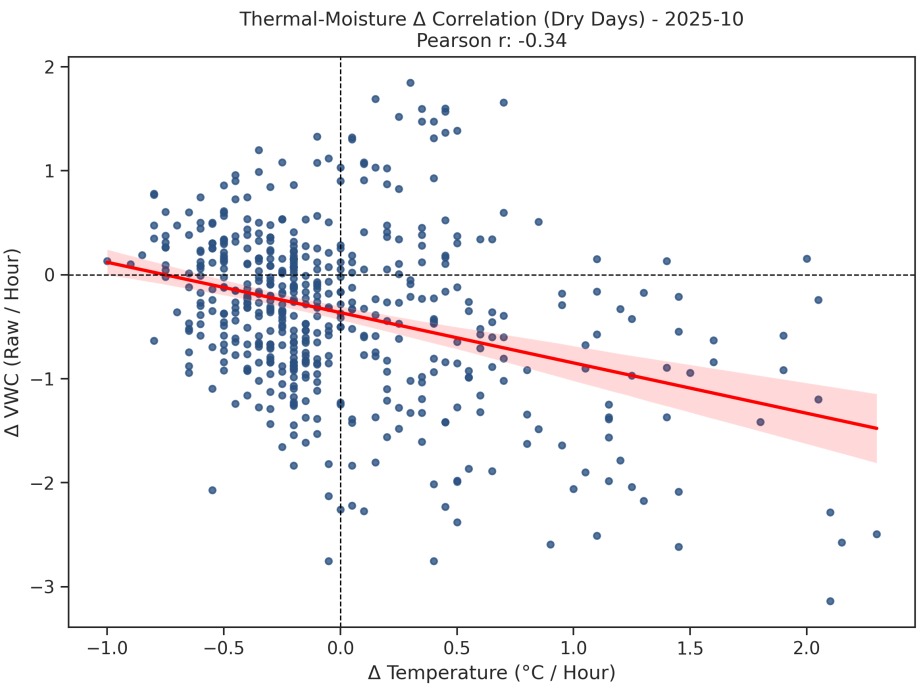
Daily VWC Distributions



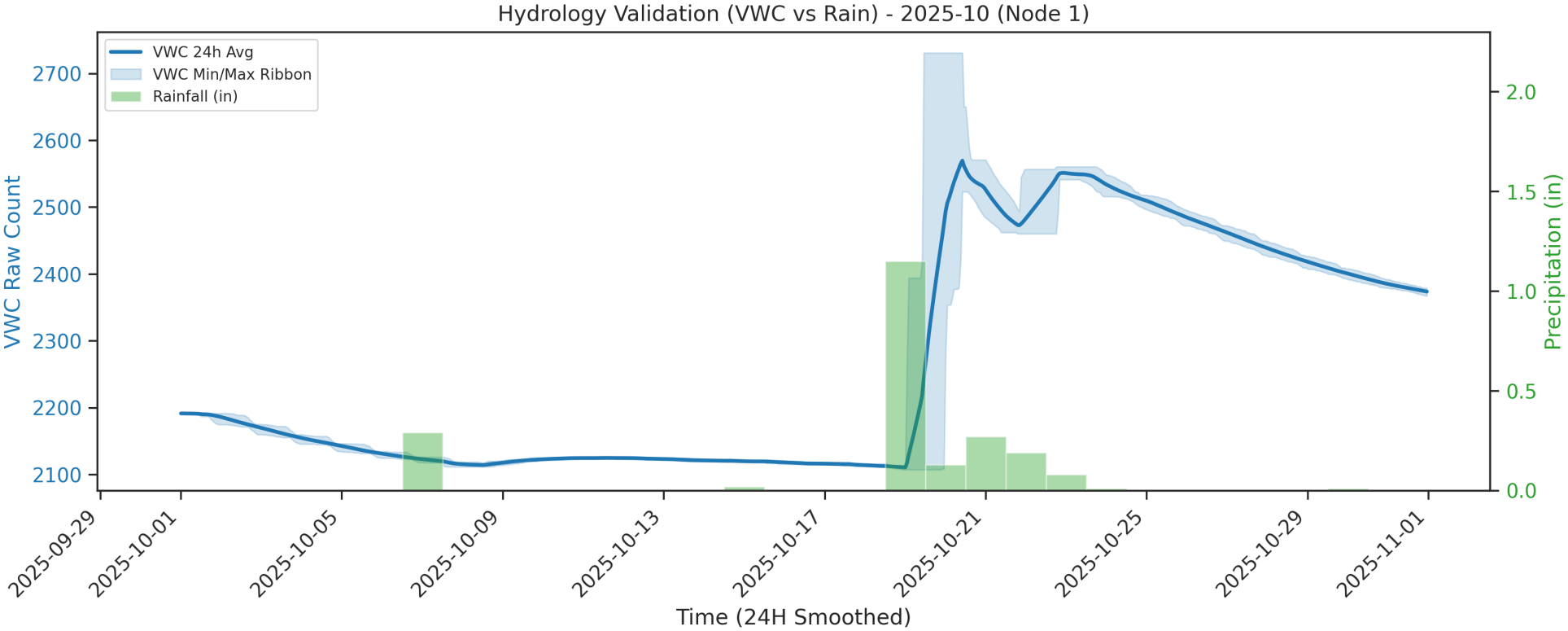
Daily boxplots showing the 25th, 50th, and 75th percentiles. Dots floating outside the whiskers are mathematically proven outliers.



24-hour rolling averages and min/max ribbons, smoothing out diurnal day/night oscillations to isolate underlying trends.



Rate of change correlation ( $\Delta$  Temp vs.  $\Delta$  VWC) on dry days, proving physical evaporation and drying curves.



Compiles raw VWC moisture counts against ground-truth weather station rainfall (green bars) to validate drainage and infiltration.